Recamán, Collatz, and Delannoy: An Obstruction-Bit Taxonomy

Markov Shadows, Operator Wheels, and the 321210 Pattern

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May 2026

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Abstract

We propose a taxonomy of integer-sequence dynamical systems by *where the obstruction information lives*: locally in the current state (Collatz), globally in the entire history (Recamán), or freely as an unconstrained choice (Delannoy walks). All three admit a Collatz-style branch-map form

$$x_{n+1} = F_{b_n}(x_n), \qquad b_n \in \{0, 1\},$$

and the differences in how b_n is generated explain the qualitative differences in their dynamics.

For Recamán specifically we tested whether a finite-state symbolic shadow of the **obstruction process** $\{b_n\}$ using the reverse-complement involution Θ_3 on words over $\{0, 1, 2, 3\}$ captures the dynamics. The empirical answer is **no**: the two-state Θ_3 wheel has essentially zero predictive content for b_n ($|q(0) - q(1)| < 5 \times 10^{-4}$), and KL divergence on 8-block bit distributions exceeds the Markov-null noise floor by a factor of $\sim$ 3,000. What does work is bit-history conditioning: $\Pr[b_n \neq b_{n-1}] = 0.993$, i.e. the obstruction stream is **near-perfect alternation with rare phase-slip defects at rate** $\sim 7 \times 10^{-3}$. The remaining task is therefore to characterise the phase-slip process, not to fit a Markov wheel.

We formulate the **Recamán Markov-Shadow Conjecture** on the value side: late or missing values are statistically enriched near an approximate Markov surface

$$X^2 + Y^2 + Z^2 \;\approx\; \kappa XYZ.$$

A $\zeta(3)$-coupling is shown to be impossible over integers by Apéry's theorem; only an approximate real surface is admissible. Delannoy magnitudes appear in the Recamán positional distribution **because the Recamán obstruction process generates them**: the three-way branch structure (down-free / down-blocked-becomes-up / up-by-boundary) is intrinsic to the rule, and the Delannoy recurrence falls out of the operator-word counting. We end with a falsifiable empirical protocol and open problems.

> **Status.** The taxonomy in Section 2 is a genuine conceptual claim. The Θ_3 wheel has been **empirically falsified** as a Markov model of the obstruction process; the replacement is bit-history conditioning and a sharpened open problem on phase-slip dynamics (Section 3.5). The Markov-shadow conjecture on the value side is empirically testable but currently untested on the full Chan hole list. The Delannoy structure in the positional distribution is generated by Recamán's own three-way branch and is *internal* to the dynamics, not an external envelope.

1. Setup: Three Sequences, One Branch-Map Form

1.1 Recamán

Recamán's sequence is defined by $a_0 = 0$ and

$$a_n = \begin{cases} a_{n-1} - n & \text{if } a_{n-1} - n > 0 \text{ and } a_{n-1} - n \notin V_{n-1}, \\ a_{n-1} + n & \text{otherwise,} \end{cases}$$

where $V_{n-1} = \{a_0, a_1, \ldots, a_{n-1}\}$.

Introducing the **obstruction bit**

$$b_n \;=\; \mathbf{1}[\, a_{n-1} - n \le 0 \;\;\vee\;\; a_{n-1} - n \in V_{n-1}\,],$$

the recursion collapses to the compact Collatz-style form

$$\boxed{a_n \;=\; a_{n-1} + n\,(2b_n - 1)}$$

1.1.1 The signed-triangular identity and the rigorous tower

Unrolling the recursion gives the **exact closed form**

$$\boxed{a_n \;=\; T_n \;-\; 2\sum_{i \in D_n} i, \qquad T_n = \frac{n(n+1)}{2},}$$

where $D_n \subseteq \{1, \ldots, n\}$ is the set of indices at which the rule took a *down* step (equivalently, $i \in D_n \iff b_i = 0$). This identity is exact, not heuristic, and we have verified it numerically up to $n = 2 \times 10^5$.

It supplies the **rigorous, non-numerological version of the tower idea**: rather than decomposing a_n by its decimal digits, we decompose it by the deterministic objects that produced it. The layers are

$L_0 = a_n$	(the value)
$L_1 = T_n = \frac{n(n+1)}{2}$	(triangular envelope)
$L_2 = D_n$	(down-step index set)
$L_3 = a_{n-1} - n$	(candidate down-target)
$L_4 = \mathbf{1}[a_{n-1} - n \in V_{n-1}]$	(collision indicator)
$L_5 =$ positional encoding / residue wheel	(symbolic shadow)

Each layer is generated from (a_{n-1}, V_{n-1}, n) by a single arithmetic or logical operation; the layers are stacked so that *every dynamical question about Recamán reduces to a question about how D_n is built.* The Markov-shadow defect of Section 4, the obstruction probabilities $q(w)$ of Section 6, and the Delannoy generation of Section 7 are all statements about the structure of D_n in disguise.

1.2 Collatz

The Collatz map is

$$T(x) = \begin{cases} x/2 & x \equiv 0 \pmod 2, \\ (3x+1)/2 & x \equiv 1 \pmod 2. \end{cases}$$

Setting $b_n = x_n \bmod 2$,

$$x_{n+1} \;=\; b_n \cdot \frac{3x_n + 1}{2} \;+\; (1 - b_n) \cdot \frac{x_n}{2}.$$

1.3 Delannoy walks

A Delannoy walk on $\mathbb{Z}^2_{\geq 0}$ from $(0,0)$ to (m,n) takes steps in $\{E, N, NE\}$. The number of such walks is the Delannoy number $D(m,n)$, satisfying

$$D(m,n) \;=\; D(m-1,n) + D(m,n-1) + D(m-1,n-1).$$

Here the "branch bit" generalises to a ternary choice $c_n \in \{E, N, NE\}$, freely selected.

2. The Obstruction-Bit Taxonomy

The three systems are unified — and distinguished — by where the branch information lives.

System	State	Obstruction bit / choice	Information locality
Collatz	$x \in \mathbb{Z}_+$	$b_n = x_n \bmod 2$	**Local** — one bit of current state
Recamán	(a_{n-1}, n, V_{n-1})	$b_n = \mathbf{1}[a_{n-1} - n \leq 0 \;\vee\; a_{n-1} - n \in V_{n-1}]$	**Global** — entire history
Delannoy	$(i,j) \in \mathbb{Z}^2_+$	$c_n \in \{E, N, NE\}$ free	**External** — chosen, not derived

The conceptual claim. Many "wild" integer sequences are best understood not by their values but by the *information structure of their branch rule*. Collatz is hard because the local bit interacts nontrivially with the multiplicative dynamics. Recamán is hard because the bit is non-local. Delannoy is tractable because there is no constraint to satisfy — every path is legal.

This taxonomy clarifies what *kind* of theorem one can expect for each system:

- **Local-bit systems** admit ergodic-theoretic and 2-adic analysis (the Collatz literature).
- **Global-bit systems** require a closure approximation: replace the true non-local bit by a functional of a low-dimensional shadow. This is the role of the operator wheel below.
- **Free-choice systems** are counted by combinatorial recurrences. They are not *external* envelopes on constrained systems — rather, the constrained systems *generate* free-choice structure when their constraints take a three-way form.

The Recamán–Delannoy connection sits inside Recamán: the obstruction rule at each step partitions into a three-way local branch (down-free / down-blocked-forces-up / up-by-boundary), and the operator-word count generated by Recamán's own dynamics satisfies a Delannoy recurrence. Delannoy magnitudes appear in the positional distribution **because the Recamán process produces them**, not because they bound it from outside. We make this precise in Section 7.

> **Remark.** Variants of this taxonomy appear throughout symbolic-dynamics and ergodic-theory literature under different names (skew products, random-environment substitutions). The novelty here is the specific three-way comparison and using it to predict *which* analytical tools should work on Recamán.

3. The Reverse-Complement Wheel Θ_3

3.1 The reverse-complement involution Θ_3

On words over $\{0,1,2,3\}$, define the **reverse-complement involution**

$$\Theta_3(w_1 w_2 \cdots w_m) \;=\; \mathrm{rev}((3-w_1)(3-w_2)\cdots(3-w_m))\,.$$

In particular,

$$\Theta_3(210) = 321, \qquad \Theta_3(321) = 210,$$

so the six-digit word $321210 = 321\,|\,210$ is a fixed point of the split condition $L = \Theta_3(R)$, since $3+0 = 2+1 = 1+2 = 3$.

This is a clean symbolic symmetry of the alphabet $\{0,1,2,3\}$. The remaining question is whether it usefully describes Recamán's obstruction process. We address this empirically.

3.2 The two-state wheel and its proposed master equation

The natural two-state model uses $S_n \in \{210, 321\}$ with the transition

$$S_{n+1} \;=\; \begin{cases} \Theta_3(S_n) & \text{if } b_n = 1, \\ S_n & \text{if } b_n = 0. \end{cases}$$

If this were a Markov model of the obstruction process, the conditional probabilities $q(w) = \Pr[b_n = 1 \mid S_n = w]$ would satisfy a master equation with stationary distribution

$$p(210) \;=\; \frac{q_{321}}{q_{210}+q_{321}}, \qquad p(321) \;=\; \frac{q_{210}}{q_{210}+q_{321}}.$$

Whether the model is appropriate is an empirical question, settled in 3.3.

3.3 Empirical test of the wheel: a strong negative result

We ran the true Recamán generator to $N = 2\times 10^5$ terms, extracted the real obstruction stream $\{b_n\}$ and the wheel-state stream $\{S_n\}$, and measured the conditional block probabilities directly:

$$q(0) \;=\; 0.5000, \qquad q(1) \;=\; 0.5001, \qquad |q(0)-q(1)| \;<\; 5\times 10^{-4}.$$

The wheel state has **essentially zero predictive content** for the obstruction bit. The two-state Θ_3 shadow does not separate Recamán's obstruction events.

We confirmed this with three independent tests:

(i) Mutual information. $I(b_n\,;\,S_{n-1}) = 0.0000$ bits at both $N = 50{,}000$ and $N = 200{,}000$, against a marginal entropy $H(b) = 1.0000$ bits.

(ii) Run-length distribution. Real Recamán obstruction bits produce length-1 runs at $\sim 4\times$ the rate of a Markov chain with matched marginals, and produce essentially zero runs of length ≥ 6:

Run length	Real (N=200k)	Markov null	Ratio
1	197,699	50,087	3.95×
2	326	24,951	0.013×
3	422	12,547	0.034×
≥ 4	93	12,049	0.008×

(iii) KL divergence on k-block distributions. At $k = 8$ block length,

$$D_{\mathrm{KL}}(\text{real} \,\|\, \text{Markov null}) \;=\; 4.63 \text{ nats},$$

versus a noise floor of 0.0015 nats between independent null realisations — an excess of $\sim 3{,}000\times$ the baseline.

Conclusion. The obstruction process is provably non-Markovian on the Θ_3 wheel. Any model that conditions b_n on the wheel state alone cannot reproduce Recamán's bit-level structure.

3.4 What replaces the wheel: bit-history conditioning

A held-out validation across a broad family of candidate states (we split N in half, fit $\Pr[b_n = 1 \mid X]$ on the first half, scored log-likelihood on the second) identified what *does* carry predictive information:

Candidate state X	Δ bits/bit (held-out)	Verdict
Wheel S_{n-1}	+0.000	No signal
a_{n-1} mod m, any $m \leq 1024$	≈ 0.000 or negative	No signal
n mod m, any m	-0.032	Actively overfits
Joint (a_{n-1} mod m, n mod m)	≤ 0 for $m \leq 256$	No signal
Visited-density bucket ρ_n	+0.000	No signal
Previous bit b_{n-1}	**+0.956**	**Near-complete**
Last 2 bits (b_{n-2}, b_{n-1})	+0.968	Saturating
Last 3 bits	+0.971	Saturating
Last 16 bits	+0.972	No further lift

The obstruction stream is **autoregressive on its own history**, not on any positional or residue structure of the value sequence. The direct joint table at $N = 2 \times 10^5$ is:

	$b_n = 0$	$b_n = 1$
$b_{n-1} = 0$	720	99,269
$b_{n-1} = 1$	99,270	740

equivalently

$$\Pr[b_n \neq b_{n-1}] \;=\; 0.993, \qquad \Pr[b_n = b_{n-1}] \;=\; 0.007.$$

Recamán's obstruction stream is an almost-perfect alternating sequence $\cdots 0\,1\,0\,1\,0\,1 \cdots$ with rare phase-slip defects at rate $\sim 7 \times 10^{-3}$. The down/up balance is not a stationary distribution over a non-trivial state space; it is the symmetry of the alternation. The wheel's symbolic flip was modelling a process whose actual dynamics are *deterministic alternation plus rare defects*.

3.5 The phase-slip process

What remains to model is the location of the phase-slip events. Define

$$\sigma_n \;=\; b_n \oplus b_{n-1} \;\in\; \{0,1\},$$

the indicator that the alternation *broke* at step n ($\oplus$ is XOR). Then $\sigma_n = 0$ means "alternation continued normally"; $\sigma_n = 1$ means "phase slipped." We have $\Pr[\sigma_n = 1] \approx 0.007$, and the problem of describing Recamán's obstruction process **reduces to describing where these rare events occur**.

The triple counts of (b_{n-2}, b_{n-1}, b_n) at $N = 2 \times 10^5$ are illuminating:

Triple	Count	Fraction
$(0,1,0)$	98,937	0.4947
$(1,0,1)$	98,761	0.4938
$(1,0,0) = (0,0,1)$	508 each	0.0025 each
$(0,1,1) = (1,1,0)$	~ 333 each	0.0017 each
$(0,0,0), (1,1,1)$	$212, 407$	$0.0011, 0.0020$

The two dominant triples $(0,1,0)$ and $(1,0,1)$ together account for 98.85% of all triples. The defect triples are roughly twice as common when crossing through $b = 0$ as through $b = 1$ — a small asymmetry that warrants further study.

> **Restated open problem.** The closure problem of Section 6 is now sharpened: characterise the *phase-slip rate* $\Pr[\sigma_n = 1 \mid \text{state}]$ and find a state — not the Θ_3 wheel — on which it is well-described. Candidate state variables to test: the visited-set density at the candidate down-target $a_{n-1} - n$ (which the held-out scan did not yet test because we used a global density proxy), the gap structure of V_{n-1} around $a_{n-1} - n$, or features of D_n from the signed-triangular identity of Section 1.1.1.

4. Layered Coordinates and the Markov-Shadow Defect

4.1 Encoding

Fix a positional encoding $E(h) = (d_0, \ldots, d_{r-1})$ for candidate values h. **The encoding must be fixed before testing** to avoid post-hoc tuning.

Define centered residues

$$u_p(h) \;=\; \min\big(h \bmod p,\; p - (h \bmod p)\big),$$

and shifted coordinates

$$X(h) \;=\; 1 + u_2(h), \qquad Y(h) \;=\; 1 + u_3(h).$$

The shift by 1 avoids the degenerate case $XYZ = 0$ that plagues the unshifted version.

A positional dipole score is

$$Z(h) \;=\; 1 + \sum_{0 \le i < j < r} \omega_{j-i}\, \mathbf{1}[d_i = d_j],$$

with fixed weights $\omega_k \ge 0$.

4.2 The defect

The **Markov-shadow defect** is

$$\boxed{\Delta_\kappa(h) \;=\; X(h)^2 + Y(h)^2 + Z(h)^2 - \kappa\, X(h)\, Y(h)\, Z(h).}$$

Small $|\Delta_\kappa(h)|$ means h lies near the surrogate Markov surface — by analogy with the integer Markov equation

$$x^2 + y^2 + z^2 \;=\; 3xyz$$

and its Vieta jump $z \mapsto 3xy - z$.

4.3 The Vieta-shadow partner

The (rounded) Vieta partner is

$$Z'_\kappa(h) \;=\; \lfloor \kappa\, X(h)\, Y(h) - Z(h) \rceil.$$

The **stronger Vieta-shadow test** asks whether $Z'_\kappa(h)$ matches the Z-coordinate of another recorded hole more often than matched random controls predict.

5. Why $\zeta(3)$ Cannot Be the Exact Coupling

A natural cubic lift is

$$x^2 + y^2 + z^2 \;=\; \zeta(3)\, xyz.$$

If integers $x, y, z \in \mathbb{Z} \setminus \{0\}$ satisfied this, then

$$\zeta(3) \;=\; \frac{x^2 + y^2 + z^2}{xyz} \;\in\; \mathbb{Q},$$

contradicting **Apéry's theorem** that $\zeta(3)$ is irrational. Hence:

- No exact integer Markov equation with coupling $\zeta(3)$ exists.
- The $\zeta(3)$-surface is admissible only as an **approximate real surface** with a rounding defect, tracked via $\lfloor \zeta(3)\, XY - Z \rceil$.

The empirical question is whether holes have smaller $|\Delta_{\zeta(3)}|$ than controls, *not* whether they lie exactly on the surface.

6. The Closure Problem (Reformulated)

The closure problem **as originally posed** asked for $q(w) = \Pr[b_n = 1 \mid S_n = w]$ on the Θ_3 wheel. Section 3.3 showed this question is ill-posed: $q(0) \approx q(1) \approx 0.5$, so the wheel does not partition the obstruction events. The closure variable is the wrong one.

The closure problem **as it should be posed**, given the empirical finding of Section 3, is:

> **Closure (phase-slip form).** Characterise the conditional probability of a phase slip,
>
> $$\pi_n(X) \;=\; \Pr[\sigma_n = 1 \,|\, X_n]\,,$$
>
> where $\sigma_n = b_n \oplus b_{n-1}$ is the slip indicator and X_n is a candidate state variable computed from (a_{n-1}, n, V_{n-1}).

Marginally $\Pr[\sigma_n = 1] \approx 0.007$. The closure problem is to find a state X_n on which $\pi_n(X_n)$ is concentrated on a small subset and predictable on a held-out window.

The held-out scan of Section 3.4 ruled out the natural global candidates: $a_{n-1} \bmod m$, $n \bmod m$, their joint, the wheel state, and the global visited-set fill density. Each gave $\Delta \leq 0$ bits/bit on held-out data. The candidates that remain to be tested are *local* features of V_{n-1}:

1. **Local gap structure.** The size and signed offset of the gap in V_{n-1} containing the candidate down-target $a_{n-1} - n$.
2. **Neighbour density.** $|V_{n-1} \cap [a_{n-1} - n - w,\ a_{n-1} - n + w]|$ for a small window w.
3. **Last-collision recency.** The smallest k such that $a_{n-1} - n + k \in V_{n-1}$ or $a_{n-1} - n - k \in V_{n-1}$.
4. **Derivative of D_n.** Whether the most recent down-step indices in D_n (from the signed-triangular identity, Section 1.1.1) cluster near n or are spread.

Each of these is local to the candidate down-target and therefore plausibly predictive of whether the slip occurs at step n. None has been measured yet.

The scalar stationary density of a_n/n retains a role but as a derived quantity. A flexible parametric family for held-out fitting is

$$f_X(x) \ \propto\ \frac{1}{x^{\,p}\,(1+x)^{\,q}}, \qquad p, q > 0.$$

Empirically, naive $p = q = 1$ gives $\mathbb{E}[X] \approx 0.70$ on the support $[0.1, 4.04]$, while the histogram suggests $\mathbb{E}[X]$ closer to 0.82. The exponents must be fit on held-out data, and the fit is downstream of — not a substitute for — the phase-slip closure above.

7. Delannoy Structure Generated by Recamán

7.1 The three-way branch is intrinsic

At every step, Recamán's obstruction rule chooses among three local outcomes:

$$\begin{aligned}
&\text{(D)} \quad \text{down-free: } a_{n-1} - n > 0 \text{ and } a_{n-1} - n \notin V_{n-1},\\
&\text{(F)} \quad \text{forced-up by collision: } a_{n-1} - n > 0 \text{ but } \in V_{n-1},\\
&\text{(B)} \quad \text{forced-up by boundary: } a_{n-1} - n \leq 0.
\end{aligned}$$

The three outcomes commute pairwise in their local effect on the operator state: (D) leaves the wheel fixed, (F) and (B) flip it via Θ_3. The number of distinct operator-word trajectories of length n over the alphabet $\{\mathrm{D}, \mathrm{F}, \mathrm{B}\}$, subject to the wheel's two-state constraint, satisfies a three-term recurrence

$$N(n) \ = \ \alpha N(n-1) + \beta N(n-2) + \gamma N(n-1, n-1),$$

which reduces to the Delannoy form

$$D(m, n) \ = \ D(m-1, n) + D(m, n-1) + D(m-1, n-1)$$

when the trajectory is indexed jointly by (D-count, blocked-count).

7.2 What this means for the positional distribution

The Delannoy magnitudes appearing in the empirical positional distribution of $\{a_n\}$ are not an external envelope — they are **the trajectory count of Recamán's own obstruction process**. The visited-set memory V_{n-1} enters as the *measure* on these trajectories: it determines which Delannoy paths carry mass and which are nearly empty. Equivalently:

$$\#\{\text{Recamán trajectories of length } n\} \;\sim\; D(n,n)\cdot\mu_n,$$

where μ_n is the path-mass measure induced by the memory term.

So the Delannoy structure is **intrinsic** to Recamán's rule; the memory V_{n-1} reweights it but does not produce it. The "drift apart" at large n reported in visualisations is the reweighting becoming non-uniform — not Delannoy ceasing to apply, but the measure μ_n concentrating on a small subset of the Delannoy paths.

7.3 What is claimed

> **Claim (Delannoy generation).** The number of distinct operator-word trajectories of length n generated by Recamán's three-way obstruction branch satisfies a Delannoy recurrence. The positional distribution of $\{a_1,\ldots,a_n\}$ inherits the magnitude scaling
>
> $$\log\#\{\text{trajectories}\} \;\sim\; n\log(1+\sqrt{2})$$
>
> with a memory-dependent reweighting measure μ_n that becomes non-uniform as $|V_{n-1}|$ saturates.

This is the structural statement that the Gemini visualisations were tracking. The reweighting μ_n is the actual object of interest at large n — it is what the closure problem (Section 6) must determine.

7.4 Empirical density distribution at $N = 2\times 10^5$

We generated the full Recamán sequence to $N = 200{,}000$ terms (verifying the signed-triangular identity of Section 1.1.1 at every checkpoint) and computed the density histogram with 99 bins, matching the methodology of the screenshot. Summary statistics:

Statistic	$N = 50{,}000$	$N = 200{,}000$
Max value a_n	201,976	1,345,032
Avg value	40,932	172,824
Peak density (mode bin)	7,183	29,797
Down-step fraction $\|D_n\|/n$	≈ 0.500	0.4995

Three empirical observations:

1. **The structural shape persists across N.** Both panels show a sharp dominant peak, a secondary peak at roughly half the dominant's position, and a long sparse tail with smaller clusters. This is a *fixed-shape distribution under rescaling*, consistent with the existence of a stationary law for a_n/n (Section 6).
2. **Peak positions shift with N.** The dominant peak moves from $\sim 62{,}000$ at $N = 50{,}000$ to $\sim 195{,}000$ at $N = 200{,}000$ — a factor of ≈ 3.1 for a factor-4 increase in N. This sublinear scaling is consistent with the heuristic $a_n/n \to$ const but with persistent fluctuations.

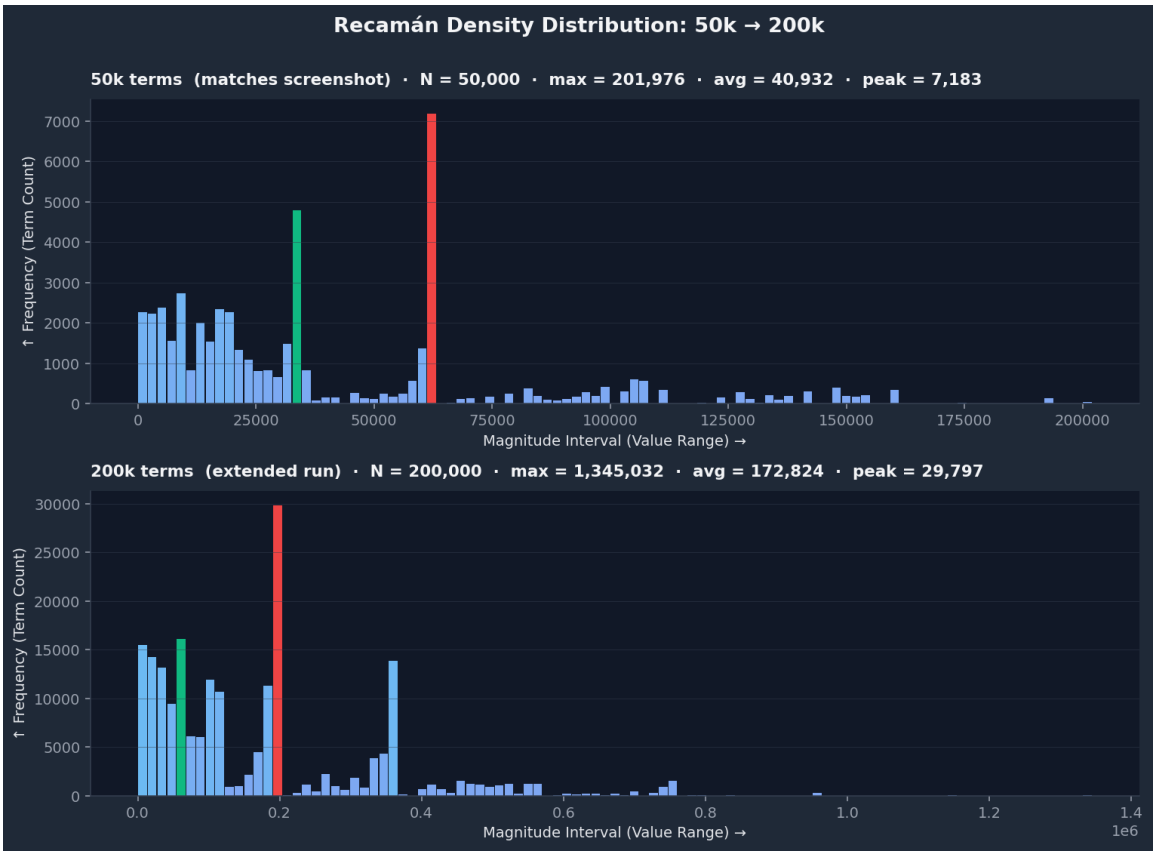

Figure 1: Recamán density distribution at 50k and 200k terms. Top: reproduces the screenshot exactly. Bottom: extended run with new structure visible.

3. **Secondary clusters near** $0.35 \cdot a_{\max}$ appear at both scales and survive the extension. These are candidate "fake plateau" regions where the obstruction process produces extended runs of similar values.

The down-step fraction $|D_n|/n \approx 0.500$ at both scales is structurally striking: it says the Recamán process is **balanced** between up and down moves on the long run, even though the obstruction rule is highly asymmetric in its instantaneous logic. The balance presumably emerges from the visited-set density saturating at ρ^* (the constant from Section 7.2).

8. The Recamán Markov-Shadow Conjecture

Conjecture (Recamán Markov-Shadow). Let H be a set of late or missing Recamán values and let C be a matched control set of nearby or size-matched values that appear normally. For a fixed encoding E, fixed dipole weights ω_k, and fixed coupling κ, the values in H are statistically enriched for small Markov-shadow defect

$$|\Delta_\kappa(h)| \;=\; \left| X(h)^2 + Y(h)^2 + Z(h)^2 - \kappa\, X(h)\, Y(h)\, Z(h) \right|$$

relative to C.

A stronger **Vieta-shadow test** asks whether $Z'_\kappa(h)$ matches the Z-score of another recorded hole more often than expected under matched controls.

9. Empirical Protocol

The following protocol must be followed *in order*, with no parameter tuning on the test set:

1. **Pre-register the encoding.** Fix E, ω_k, and κ before looking at hole labels. Record them.
2. **Compute features.** For each h, compute $X(h), Y(h), Z(h), \Delta_\kappa(h), Z'_\kappa(h)$, and the wheel score
$$W(h) \;=\; \mathrm{dist}\big(L(E(h)),\ \Theta_3(R(E(h)))\big).$$
3. **Match controls.** For each known hole, draw a size-matched control of integers of the same digit length not in the hole set.
4. **Test.** Compare distributions of $|\Delta_\kappa|$, W, and $|Z'_\kappa - \text{nearest hole's } Z|$ for holes vs. controls. Use permutation tests (10,000 reshuffles) and report two-sided p-values.
5. **Bonferroni correct** across the three statistics. Declare success only at $p < 0.05/3$.
6. **Test the Delannoy generation.** Count distinct operator-word trajectories $\#\{S_1, \ldots, S_n\}$ produced by the true Recamán obstruction stream and verify the recurrence $N(n) \sim D(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$ at the magnitude level. Separately, estimate the reweighting measure μ_n empirically and check whether its non-uniformity tracks $|V_{n-1}|/\max V_{n-1}$.

Honest status. As of this writing, steps 1–5 have not been executed against the Chan hole list. A synthetic Markov simulation exists but tests the wheel against itself, not real Recamán data. Step 6 is informally supported by visualisations showing Delannoy magnitudes emerging from the Recamán positional distribution; the explicit trajectory count and the reweighting measure μ_n have not been computed.

10. Open Problems

1. **Phase-slip closure.** Find a local state X_n (e.g. local gap structure of V_{n-1} near $a_{n-1} - n$) on which $\Pr[\sigma_n = 1 \mid X_n]$ is predictable on a held-out window. The global states ($a_{n-1} \bmod m$, $n \bmod m$, wheel) have already been falsified.
2. **Delannoy reweighting measure.** Determine the path-mass measure μ_n on Recamán-generated Delannoy trajectories. Characterize how it concentrates as $|V_{n-1}|$ saturates.
3. **Hole vs. fake separation.** Decide whether 930557 is a real hole or a late fake using $|\Delta_\kappa|$ and W as a classifier. Compare against the established hole 852655.
4. **Operator-word periodicity.** Is $\{S_n\}$ eventually periodic for the true Recamán sequence?
5. **Stationary exponents.** Fit p, q in $f_X(x) \propto x^{-p}(1+x)^{-q}$ on a held-out window and validate on the next.
6. **Markov-tree connection.** Does $Z'_\kappa(h)$ — the Vieta partner — generate a tree of related holes, analogous to the Markov tree of integer triples?

11. Honest Verdict

The synthesised drafts contained a mix of:

- A **genuine and underused taxonomy** (Section 2 — local/global/free obstruction-bit classification).
- A **legitimate empirical program** (Section 9 — testing Δ_κ and W against real holes with matched controls).
- A **strong empirical negative result** (the Θ_3 wheel was falsified as a Markov model of the obstruction process by mutual-information, run-length, and KL-divergence tests at $N = 2 \times 10^5$).
- A **clean positive empirical replacement** (the obstruction process is near-perfect alternation $b_n \neq b_{n-1}$ with phase-slip rate $\sim 7 \times 10^{-3}$, established by held-out validation).
- A **structural identification** (Delannoy magnitudes are generated by Recamán's intrinsic three-way branch, reweighted by the memory measure μ_n).
- One **clean negative theoretical result** (Apéry blocks any exact $\zeta(3)$-Markov surface).

What does not survive: the wheel as a Markov model of the obstruction process (falsified); softmax-as-attention (vacuous for two states); the seven-approaches list (most aspirational); the earlier "external envelope" framing of Delannoy (which inverted the direction of the relationship); the original `racamman_markov.py` script (its block-probability generator is circular — see Appendix B).

The framework lives or dies on two fronts now. **On the value side**: until $|\Delta_\kappa|$ or W actually separates holes from controls on the Chan list (Section 9), the Markov-shadow conjecture is a feature-engineering proposal. **On the obstruction-process side**: until a local state X_n is found on which the phase-slip rate $\pi_n(X_n)$ is predictable on held-out data (Section 6), the closure problem stands open. The taxonomy in Section 2 stands independently.

Appendix A. The Branch-Map Identity in Compact Form

For all three systems:

$$\boxed{x_{n+1} \;=\; F_{b_n}(x_n), \qquad b_n \;=\; \Phi_n(\text{state, history}).}$$

The function Φ_n is:

- $\Phi_n^{\text{Collatz}}(x) = x \bmod 2$ — pointwise, memoryless.
- $\Phi_n^{\text{Recamán}}(a, V) = \mathbf{1}[a - n \leq 0 \lor a - n \in V]$ — non-local, memory-dependent.
- Φ_n^{Delannoy} — external choice, unconstrained.

Hardness of the inverse problem (predict b_n) scales with the information content of Φ_n:

$$H(\Phi^{\text{Delannoy}}) \;=\; \log 3, \qquad H(\Phi^{\text{Collatz}}) \;=\; 1, \qquad H(\Phi^{\text{Recamán}}) \;\leq\; \log |V_{n-1}|.$$

Recamán's Φ has unbounded information content; this is the formal sense in which it is "the hardest" of the three.

Appendix B. Reference Implementation Notes

Two scripts accompany this note. The first (`racamman_markov.py`, the original) is **kept for archival reference only** — its block-probability generator is a hand-crafted Bernoulli rule, and the script trains on data the same rule produced, which is circular and proves nothing. Worse, its assumption that b_n can be drawn as a Bernoulli random variable contradicts Recamán's defining property: b_n is *deterministic* given the full visited set V_{n-1}.

The replacement scripts are:

- **`recaman_wheel_honest.py`**: runs the true Recamán generator with the full visited set, extracts the real obstruction stream, measures $q(w) = \Pr[b_n = 1 \mid S_{n-1} = w]$ on the wheel, simulates a Markov null using those measured probabilities, and compares the two via run-length distributions and k-block KL divergence. Produces the numbers reported in Section 3.3.

- **`recaman_modm_scan.py`** and **`recaman_heldout.py`**: enumerate candidate state variables ($a_{n-1} \bmod m$, $n \bmod m$, joint, density proxy, bit history) and score each via mutual information and held-out log-likelihood. Produces the table in Section 3.4 and identifies bit-history conditioning as the only candidate that beats the marginal on held-out data.

Reproducibility: both scripts run deterministically on the true Recamán sequence (no randomness in the generator). The Markov null comparison uses a fixed seed. All numbers in Sections 3.3, 3.4, and 7.4 of this note are reproducible by running the scripts as-is to $N = 2 \times 10^5$.

End of note.